



UNIVERSITY OF SANTO TOMAS
COLLEGE OF INFORMATION AND COMPUTING SCIENCES
DEPARTMENT OF INFORMATION TECHNOLOGY
Academic Year 2025–2026 | Special Term



SYSTEM USABILITY SCALE (SUS) EVALUATION INSTRUMENT

ISO/IEC 25010 Usability & User Experience Quality Metric

Project Title: SolanaWatch: An IoT-Based Disease Early Warning System for Soil-Grown Solanaceae Crops

Proponents (3ITC - Group 1): • Morales, Simeon Agabus S. • Rapada, Raven M. • Hernandez, Andrei S. • Eugenio, Vince Angelo	Academic Supervision & Metric: • Project Adviser: Prof. Eugenia R. Zhuo, DIT • Degree: B.S. in Information Technology • Target Usability Score: ≥ 80.0 / 100 (Grade A) • Target Users: Smallholder Farmers & Technicians
Participant ID / Code: _____	Date of Evaluation: _____

SECTION 1: PURPOSE OF EVALUATION & INFORMED CONSENT

Dear Participant,

Thank you for participating in the usability evaluation of **SolanaWatch: An IoT-Based Disease Early Warning System for Soil-Grown Solanaceae Crops**. SolanaWatch is a capstone research system designed at the University of Santo Tomas – CICS. It combines solar-powered field sensor nodes (capturing air temp, relative humidity, leaf wetness duration, root-zone soil temp, and soil moisture) with a responsive web dashboard to provide pre-symptomatic alerts for Late Blight (*Phytophthora infestans*) and Bacterial Wilt (*Ralstonia pseudosolanacearum*).

This evaluation utilizes the industry-standard **System Usability Scale (SUS)** (Brooke, 1996) to measure usability, learnability, and overall user acceptance under ISO/IEC 25010 standards. Your honest assessment will guide final technical refinements.

Informed Consent & Data Privacy Guarantee (R.A. 10173 - Data Privacy Act of 2012):

- Participation is entirely voluntary. You may pause or withdraw from this evaluation at any time without penalty.
- All responses will be anonymized and kept strictly confidential using assigned Participant IDs.
- Data collected will be used exclusively for academic research, capstone documentation, and scientific dissemination.

SECTION 2: PARTICIPANT DEMOGRAPHIC & OPERATIONAL PROFILE

Please mark [X] or fill in the information that best describes your background:

1. Primary Stakeholder Role: ☐ Smallholder Solanaceae Farmer ☐ Agricultural Extension Worker / LGU Technician ☐ Agronomist / Plant Pathologist ☐ Academic Researcher / IT Evaluator ☐ Other: _____	2. Gender & Age Bracket: Gender: ☐ Male ☐ Female ☐ Prefer not to say Age: ☐ 18–29 ☐ 30–39 ☐ 40–49 ☐ 50–59 ☐ 60 and above
3. Farming / Agricultural Experience: ☐ Less than 2 years ☐ 2 to 5 years ☐ 6 to 10 years ☐ More than 10 years	4. Solanaceae Crops Cultivated / Handled: ☐ Eggplant (Talong) ☐ Tomato (Kamatis) ☐ Pepper (Siling Haba / Bell Pepper) ☐ Potato (Patatas) ☐ Mixed / Diversified Cropping
5. Farm / Workplace Location: Barangay: _____ Municipality: _____ Province: _____	6. Digital Device Usage in Daily Routine: ☐ Smartphone (Android / iOS) ☐ Tablet ☐ Desktop / Laptop Computer ☐ Basic Keypad Phone (SMS only) ☐ Non-User / Assistance Required
7. Prior Experience with Smart Farming Tools: ☐ None (First time using IoT farm tools) ☐ Occasional (Weather apps, SMS advisories)	8. Preferred Language for Farming Advice: ☐ English ☐ Filipino / Tagalog

☐ Frequent (Sensors, telemetry dashboards)	☐ Bilingual (English with Filipino subtitles) ☐ Regional Dialect: _____
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SECTION 3: SYSTEM INTERACTION & TASK WALKTHROUGH

Before completing the questionnaire, please ensure you have interacted with the SolanaWatch Web Application by performing the five core user scenarios:

Task #	Operational Scenario	Core User Action Performed	Completion
Task 1	Live Telemetry Inspection	Navigated dashboard to inspect live air temp, air RH, leaf wetness duration, root-zone soil temp, and soil moisture gauges.	☐ Done ☐ Partial
Task 2	Disease Early Warning Alerts	Observed and interpreted color-coded alert banners (Green = Low, Yellow = Moderate, Red = High Risk) for Late Blight and Bacterial Wilt.	☐ Done ☐ Partial
Task 3	Historical Trend Analysis	Interacted with historical time-series charts to inspect 24-hour diurnal microclimate swings and prolonged leaf wetness events.	☐ Done ☐ Partial
Task 4	Actionable Agronomic Advice	Accessed and reviewed specific farm management advisories triggered by active risk levels (e.g., canopy thinning, drainage trenching).	☐ Done ☐ Partial
Task 5	Hardware Health Telemetry	Checked system health telemetry, including LiFePO4 battery percentage, MPPT solar charging state, and offline MicroSD sync status.	☐ Done ☐ Partial

SECTION 4: SYSTEM USABILITY SCALE (SUS) QUESTIONNAIRE

Instructions: Please mark [X] in the box that corresponds to your immediate, honest rating for each statement.

Tagalog: Mangyaring lagyan ng [X] ang kahon na pinakaangkop sa inyong karanasan sa paggamit ng sistema.

Scale: 1 = Strongly Disagree (SD) | 2 = Disagree (D) | 3 = Neutral (N) | 4 = Agree (A) | 5 = Strongly Agree (SA)

#	Usability Statement (English & Filipino)	1 SD	2 D	3 N	4 A	5 SA
1	I think that I would like to use the SolanaWatch system frequently. <i>Nais kong gamitin ang SolanaWatch system nang madalas sa aking sakahan/trabaho.</i>	☐	☐	☐	☐	☐
2	I found the system unnecessarily complex. <i>Pakiramdam ko ay masyadong masalimuot o kumplikado ang sistema nang walang sapat na dahilan.</i>	☐	☐	☐	☐	☐
3	I thought the system was easy to use. <i>Pakiramdam ko ay madaling gamitin at unawain ang sistema.</i>	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to be able to use this system. <i>Sa tingin ko ay kakailanganin ko ang tulong ng isang eksperto o teknikal na tao upang magamit ang sistemang ito.</i>	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated. <i>Napansin kong maayos na magkakaugnay at nagtutulungan ang iba't ibang bahagi at impormasyon sa sistema.</i>	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system. <i>Pakiramdam ko ay labis ang hindi pagtutugma o pabagu-bago sa sistema.</i>	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly. <i>Tinataya kong karamihan ng mga magsasaka at gumagamit ay mabilis na matututong gumamit ng sistemang ito.</i>	☐	☐	☐	☐	☐
8	I found the system very cumbersome (awkward/heavy) to use. <i>Naramdaman kong napakahirap, nakakaasiwa, o mabigat gamitin ang sistema.</i>	☐	☐	☐	☐	☐

9	I felt very confident using the system. <i>Naging lubos akong tiwala at panatag sa paggamit ng sistema.</i>	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system. <i>Kinailangan kong mag-aral ng napakaraming bagay bago ko nagawang gamitin nang maayos ang sistema.</i>	☐	☐	☐	☐	☐

SECTION 5: QUALITATIVE USABILITY FEEDBACK & SUGGESTIONS

Please share brief qualitative remarks to guide future software and hardware enhancements:

1. What specific dashboard features, environmental charts, or alerts were the MOST HELPFUL and easiest to understand?

Anong partikular na bahagi ng dashboard, grap, o babala ang pinakanakatulong at pinakamadaling maunawaan?

2. Did you encounter any CONFUSION, difficulty, or visual clutter while navigating or reading disease risk levels?

May naging kalituhan ba, kahirapan, o magulong impormasyon habang tinitingnan ang antas ng peligro ng sakit?

3. What ENHANCEMENTS or additional features would you recommend for real-world deployment on your farm?

Anong mga dagdag na kakayahan, pagbabago sa wika, o disenyo ang iminumungkahi ninyo para sa aktwal na sakahan?

SECTION 6: EVALUATOR SCORING GUIDE & USABILITY BENCHMARKS (AUDIT REFERENCE)

Note for Capstone Panel & Evaluator: The System Usability Scale produces a composite score from 0 to 100 points (Brooke, 1996). Follow the standardized calculation below:

Brooke (1996) Scoring Formula: 1. Odd Items (1, 3, 5, 7, 9): Contribution = Scale Position – 1 2. Even Items (2, 4, 6, 8, 10): Contribution = 5 – Scale Position 3. Composite Normalized Score: SUS Score = [Sum of all 10 item contributions] × 2.5 Range: 0 to 100 points.	Evaluator Official Score Card: - Sum of Odd Contributions (X): _____ - Sum of Even Contributions (Y): _____ - Total Raw Sum (X + Y): _____ Calculated Composite SUS: _____ / 100 - Adjective Rating: _____ - ISO/IEC Usability Target Met: [] YES [] NO
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SUS Score Interpretation Scale (Bangor et al., 2008; Sauro & Lewis, 2012):

SUS Score Range	Letter Grade	Adjective Rating	Acceptability Status	SolanaWatch Compliance
80.3 – 100.0	Grade A	Excellent / Best Imaginable	Highly Acceptable	★ Target Design Goal (≥ 80.0)
68.0 – 80.2	Grade B / C+	Good (Above Average)	Acceptable (Industry Avg = 68.0)	Meets Baseline Standard
51.0 – 67.9	Grade C / D	OK / Marginal	Marginally Acceptable	Requires UI Revisions
0.0 – 50.9	Grade F	Poor / Unacceptable	Not Acceptable	Critical Usability Failure

SECTION 7: PARTICIPANT ACKNOWLEDGEMENT & EVALUATOR SIGN-OFF

Declaration: I hereby confirm that I have interacted with the SolanaWatch system and completed this evaluation truthfully and voluntarily.

Evaluated by (Participant):_____
Signature over Printed Name

Date Signed: _____

Administered & Verified by:_____
Lead Usability Evaluator (UST-CICS)

Date Verified: _____